

NOTICE OF INTENT to Sole Source

1. **SOLICITATION NUMBER:** NIMH25001453_1
2. **TITLE:** Fieldline 64 Sensor
3. **CLASSIFICATION CODE:** R499 – Support – Professional: Other
NAICS CODE: 334510 – Electromedical and Electrotherapeutic Apparatus Manufacturing.
4. **RESPONSE DATE:** March 18, 2025, at 12:00 PM EST
5. **PRIMARY POINT OF CONTACT:**
Joe Kennedy
Joseph.kennedy@nih.gov
Phone: 262-391-7630
6. **DESCRIPTION:**

INTRODUCTION

This is a notice of intent, not a request for a quotation.

THIS IS A PRE-SOLICITATION NON-COMPETITIVE (NOTICE OF INTENT) SYNOPSIS TO AWARD A CONTRACT WITHOUT PROVIDING FOR FULL OR OPEN COMPETITION (INCLUDING BRAND-NAME).

The National Institute of Mental Health (NIMH) intends to negotiate on a sole source basis with Fieldline Industries, Inc., for a 12-month base contract.

NORTH AMERICAN INDUSTRY CLASSIFICATION SYSTEM (NAICS) CODE and SET ASIDE STATUS

The intended procurement is classified under NAICS code 334510 – Electromedical and Electrotherapeutic Apparatus Manufacturing with a Size Standard of 1,250 Employees.

This intended procurement will be a small business set-aside requirement.

REGULATORY AUTHORITY

This acquisition is conducted under the authority of the Federal Acquisition Regulation (FAR) Part 13—Simplified Acquisition Procedures. Contracts awarded using FAR Part 13—Simplified Acquisition Procedures are exempt from the requirements of FAR Part 6—Competition Requirements. FAR Subpart 13.106-1 (b) Soliciting from a single source provides that: For purchases not exceeding the simplified acquisition threshold, Contracting Officers may solicit from one source IF the contracting officer determines that the circumstances of the contract action deem only one source reasonably available (e.g., urgency, exclusive licensing agreements, brand-name OR industrial mobilization).

The resultant contract will include all applicable provisions and clauses in effect through the Federal Acquisition Circular (FAC) Number 2024-05, dated April 22, 2024.

DESCRIPTION OF REQUIREMENT

Purchase Description

Currently, non-invasive monitoring of human brain function is limited to high spatial resolution/low temporal resolution techniques such as magnetic resonance imaging (MRI), or low spatial resolution/high temporal resolution techniques such as magnetoencephalography (MEG) or electroencephalography (EEG). There are currently no non-invasive techniques that can measure human brain function with both high spatial and temporal resolution. Thus, the only option for high temporal and spatial resolution mapping of neuronal activity is invasive electrocorticography (ECoG), which is only performed in situations such as presurgical planning for epilepsy due to inherent risks. As a result, most of the knowledge neuroscientists have obtained from ECoG studies have been in abnormal brains. The development of a non-invasive high temporal and spatial resolution array could transform the treatment of epilepsy and human neuroscience.

Recently, significant development has occurred in the MEG field in the use of optically pumped magnetometer (OPM) arrays. These relatively portable arrays are placed directly on the scalp and do not require cooling with expensive and increasingly scarce liquid helium. Most groups have focused on the development of whole-head wearable arrays that allow naturalistic movement of participants or facilitate whole brain mapping in young children. The array we propose will utilize a significantly denser array of sensors, capable of mapping brain function with dramatically improved spatial resolution – performing magnetocorticography (MCoG) as an alternative to ECoG. While we have constructed an initial prototype, it has proved inadequate to fully demonstrate the feasibility of MCoG, and additional testing with a more suitable prototype is warranted. Our generation-1 prototype device was created using standard OPM sensors; we propose a generation-2 prototype with dual-axis sensors that will demonstrate feasibility of the MCoG technique. Following this demonstration, we propose designing an MCoG optimized OPM sensor, designed specifically for high-resolution, precision mapping. We propose a novel compound sensor, incorporating both multiple cells and a reference cell for gradiometry. In order to complete the generation-2 prototype, we require the next generation of OPM sensors (Fieldline V3 sensors). In addition, in pursuit of our larger aim of developing a novel MCoG optimized sensor, we require research and design services to evaluate preliminary feasibility of a compound gradiometer.

Specific Requirements

The objective of the current project is to build and validate a non-invasive magnetocardiography (MCoG) array that can collect data at high temporal and spatial resolution, dramatically increasing our ability to observe human neurophysiology non-invasively, potentially transforming our study of the brain.

In the initial phase we will simulate and build a prototype array using currently available dual-axis OPMs, demonstrate the feasibility of MCoG, and begin work on the design of an optimized OPM.

To achieve this end, the awarded contractor will help with the following objectives:

Objective 1: Provide hardware, to include 64 Fieldline V3 sensors, electronics chassis and peripherals, sensor electronics, and an analog input card.

Objective 2: Provide research and design services including laser selection and prototyping of a compound gradiometer.

Salient Characteristics:

To perform the work, the contractor must meet the following criteria:

- Have the instrumentation and tools necessary for the research and development of our OPM magnetoencephalography systems including:
 - optically-pumped magnetometers (OPMs),
 - custom electronics
 - wearable helmets
 - other related technologies.
- Have design, prototyping, and manufacturing capabilities including:
 - Vacuum systems for vapor cell and thermal packaging R&D
 - Custom MEMS cell fabrication equipment
 - Clean rooms and hoods
 - Lasers and optics lab with related tools and equipment
 - Electronics lab with tools and equipment for prototyping and testing
 - A full suite of electronic components and assembly tools
 - Optical Microscopes
 - Various multi-layer magnetic shields
 - Calibration coils and drivers
 - Fluxgate magnetometers
 - Spectrometers
 - Oscilloscopes

- Arbitrary waveform generators
- Lock-in amplifiers
- Low-noise power supplies
- Optical power meter
- Data acquisition systems
- 3D printers for prototyping and manufacturing
- UV and thermal curing stations
- Various electronic and sensor testing equipment

Delivery Date

30 days after receipt of purchase order.

Period of Performance

The proposed period of performance shall be for one year.

Base: 4/1/2025 – 3/31/2026

Government Responsibilities.

There will be no Government Furnished Equipment as part of this Purchase Order. The PI at NIMH will oversee the progress, as demonstrated in monthly meetings and annual reports.

Data Rights.

Software used by the contractor in the fulfillment of Objective 2 will be controlled by the contractor.

Delivery or Deliverables.

- 64 zero-field optically-pumped magnetometers (OPMs)
- 19" chassis 3U for sensor cards and peripherals, V3
- Sensor electronics, each controlling 4 OPMs, V3
- Analog input card for 4 bi-polar analog inputs, BNC
- Research and design services including laser selection and prototyping of a compound gradiometer

Section 508—Electronic and Information Technology Standards

The contractor shall comply with Section 508 of the Rehabilitation Act (29 U.S.C. § 794d), as amended by the Workforce Investment Act of 1998 (P.L. 105-220), August 7, 1998.

(INCLUDING BRAND-NAME) DETERMINATION

The determination by the Government to award a contract without providing for full and open competition is based upon the market research conducted. Specifically, Fieldline

Industries Inc., is the only vendor in the marketplace that can provide the services required by National Institute of Mental Health.

The intended source is:

Fieldline Industries, Inc.
4865 Sterling Drive
Boulder CO 80301

The FieldLine V3 sensors, electronics chassis and peripherals, sensor electronics and analog input card along with the research and design services for laser selection and prototyping of a compound gradiometer listed herein is available from only one source, and competition is precluded for the reasons indicated below. There are no substitutes available.

The results of the market research indicate that the essential characteristics of the sensors and accessories limit the availability to a sole source. The selected vendor, FieldLine, is the only company who can provide the design of a densely packed OPM arrays. Our facility has been using FieldLine for over five years. During our market research for the OPM sensors, there were only two viable vendors: FieldLine and QuSpin. FieldLine and QuSpin were started by the same group, so there were a lot of similar underpinnings. Other groups capable of providing OPMs were not developing sensors for biomedical applications.

CLOSING STATEMENT

THIS SYNOPSIS IS NOT A REQUEST FOR COMPETITIVE PROPOSALS. However, interested parties may identify their interest and capability to respond to this notice.

A determination by the Government not to compete this proposed contract based upon responses to this notice is solely within the discretion of the Government. All responsible sources may submit a capability statement, proposal, or quotation which shall be considered by the agency. The information received will normally be considered solely for the purposes of determining whether to proceed on a non-competitive basis or to conduct a competitive procurement.

Responses to this notice shall contain sufficient information to establish the interested parties' bona-fide capabilities for fulfilling the requirement and include: unit price, list price, shipping and handling costs, the delivery period after contract award, the prompt payment discount terms, the F.O.B. Point (Destination or Origin), the Unique Entity ID UEI), the Taxpayer Identification Number (TIN), and the certification of business size.

Responses must include the place of manufacturing (i.e. address if supply/equipment is a domestic end product and include country of manufacture). All offerors must have an active registration in the System for Award Management (SAM) www.sam.gov.

All responses must be received by the closing date and time of this announcement and must reference the solicitation number, NIMH25001453_1.

Responses must be submitted electronically to Joe Kennedy, Contract Specialist, at joseph.kennedy@nih.gov

U.S. Mail and fax responses will not be accepted.